

# Dissecting NSEC and NSEC3 Deployment in the Wild

Yizhe Zhang\*

University of Virginia  
Charlottesville, Virginia, USA  
yz6me@virginia.edu

Shumon Huque

Salesforce  
McLean, Virginia, USA  
shuque@gmail.com

Hyeonmin Lee\*

Hanyang University  
Ansan, Gyeonggi-do, South Korea  
hyeonminlee@hanyang.ac.kr

Yixin Sun

University of Virginia  
Charlottesville, Virginia, USA  
ys3kz@virginia.edu

## Abstract

Authenticated denial of existence is a fundamental component of DNS Security Extensions (DNSSEC), realized through NSEC and NSEC3 records. Over time, several variants have emerged to improve the efficiency and privacy of denial-of-existence responses, including Compact Denial of Existence. Despite their operational importance, the prevalence and diversity of NSEC and NSEC3 deployments in the wild remain poorly understood. In this paper, we present a comprehensive measurement study of NSEC and NSEC3 deployment across 174 million domains in the .com, .net, and .org zones, as well as domains from four popular rankings. A key challenge in such measurement lies in accurately distinguishing among NSEC/NSEC3 variants across heterogeneous provider implementations. To address this challenge, we develop novel metrics driven by statistical observations, which reveal prevalent NSEC/NSEC3 variants such as Compact Denial of Existence in NSEC and White Lies in NSEC3. We also show that wildcard synthesis is non-negligible in the wild (potentially affecting nearly 19% and 15% of NSEC-enabled and NSEC3-enabled domain in .com, respectively), and its interaction with denial-of-existence proofs can reveal structural information of DNS zone. In particular, the transition from NODATA to NXDOMAIN responses suggest the presence of closest enclosers and enable inference of partial zone structure.

## CCS Concepts

• **Networks** → **Naming and addressing**; **Network measurement**.

## Keywords

Measurement, DNS, DNSSEC, NSEC, NSEC3

### ACM Reference Format:

Yizhe Zhang\*, Hyeonmin Lee\*, Shumon Huque, and Yixin Sun. 2026. Dissecting NSEC and NSEC3 Deployment in the Wild. In *Proceedings of the 2026 ACM Internet Measurement Conference (IMC '26)*, October 12–16, 2026, Karlsruhe, Germany. ACM, New York, NY, USA, 16 pages. <https://doi.org/10.1145/3777912.3839807>

\* Both Yizhe Zhang and Hyeonmin Lee contributed equally to this work.



This work is licensed under a Creative Commons Attribution 4.0 International License. *IMC '26, Karlsruhe, Germany*

© 2026 Copyright held by the owner/author(s).  
ACM ISBN 979-8-4007-2327-8/2026/10  
<https://doi.org/10.1145/3777912.3839807>

## 1 Introduction

The Domain Name System (DNS) was designed without built-in security guarantees, making it vulnerable to various attacks such as cache poisoning and spoofing [7, 22, 31]. Beyond these threats, DNS has another fundamental challenge: how to securely prove that a queried domain name or record type does not exist. This challenge is critical because forged negative responses could falsely indicate that existing names or records does not exist.

DNS Security Extensions (DNSSEC) [33–35] address these issues by providing data integrity, origin authentication, and authenticated denial of existence. Specifically, DNSSEC introduces NextSECure (NSEC) [33–35] and NSEC3 records [3, 19] to support authenticated denial of existence. While NSEC provides signed proofs of non-existence by chaining plaintext domain names in canonical order, it enables zone enumeration [36]. NSEC3 mitigates this by hashing domain names and organizing proofs in the hashed namespace, at the cost of additional parameter complexity.

Over time, several mechanisms have been proposed to further improve the privacy and efficiency of NSEC/3, which we use to collectively refer to both NSEC and NSEC3. These include *minimally covering NSEC* [40] and *White Lies* [17], which synthesize NSEC/3 ranges that do not reveal the actual neighboring names in the zone, and *Compact Denial of Existence (CDE)* [21], which reduces the response overhead of negative answers by encoding NXDOMAIN as a compact NODATA response.

Despite its security importance, NSEC/3 deployment has only been studied by very few prior works [13, 15]. However, none provides a unified and up-to-date view of the broader NSEC/3 ecosystem: prior work either focuses on early adoption trends [15] or narrows its scope to NSEC3 parameters [13]. Moreover, since the first study in 2019, the DNSSEC deployment landscape has changed substantially; DNSSEC adoption in .com alone has grown from 911K signed-domains reported in 2019 [15] to 8.3M signed-domains (in our study). This growth motivates a fresh measurement of how NSEC/3 deployment and its variants have evolved alongside broader DNSSEC adoption.

Furthermore, accurate measurement has become more challenging as newer NSEC/3 variants [17, 21, 40] have been standardized and deployed. These variants can synthesize denial-of-existence responses in provider-specific ways, as the relevant specifications define the high-level behavior but leave room for different concrete constructions. As a result, simple response code checks are insufficient. This complexity is further compounded by wildcard synthesis.

Despite its critical role in DNS resolution, no prior work has conducted a deep-dive analysis into how wildcard-enabled domains interact with the denial-of-existence ecosystem.

In this paper, we present a comprehensive measurement study of NSEC/3 deployment in the wild. We analyze domains from .com, .net, and .org zone files as well as four popularity-based rankings, covering 174M domains in total. Our study combines large-scale DNS measurements, active subdomain probing, and heuristic-based variant identification to characterize NSEC/3 adoption and deployment variants, while also examining wildcard behavior as an operational signal that can reveal partial zone-structure information.

**Contributions.** Our key findings and contributions are:

- We design and implement a *measurement framework* for characterizing authenticated denial-of-existence deployment at Internet scale. Our framework combines OpenINTEL datasets, active subdomain probing, and heuristic-based analysis to measure NSEC/3 variants adoption, and wildcard synthesis across 174M domains.
- We show that authenticated denial-of-existence mechanisms have grown substantially since prior measurements. In .com, for instance, the number of domains deploying NSEC increased from roughly 148K (in 2019) to 1.5M (in our snapshot), while NSEC3 increased from 744K to 6.7M. We also find a clear split between domain populations: NSEC3 dominates in the .com, .net, and .org zones, whereas NSEC is more common in popularity-based rankings.
- We introduce a novel methodology to distinguish NSEC/3 variants across heterogeneous provider implementations. Our analysis reveals that *CDE* dominates NSEC deployments, affecting more than 40% of NSEC-enabled domains in .com, .net, and .org, whereas *CDE* is rarely observed in NSEC3 deployments, where *White Lies* accounts for 6-12% of domains.
- We present the first systematic study of wildcard synthesis in DNSSEC denial-of-existence deployments. We show that wildcard-related candidates cover about 20% of NSEC-enabled domains across our datasets and that wildcard synthesis interacts can reveal partial structural information about DNS zones.

**Artifacts.** We release our measurement framework and analysis scripts in a public repository [41].

## 2 Background

We first provide background on NSEC/3 and its variants, focusing on server-side deployment (rather than client-side behavior such as resolver support).

**NSEC and NSEC3.** DNSSEC provides authenticated denial of existence using NSEC and NSEC3 records, preventing attackers from forging negative responses to spoof non-existence. NSEC [33–35] constructs such proofs by linking existing names in canonical order. Each NSEC record specifies its owner name, the next existing name in the zone, and a type bitmap identifying the record types present at the owner name. An example is shown below.

| # Owner name                 | # Next domain name | # Type bitmap |
|------------------------------|--------------------|---------------|
| a.nsec-noext.com. 60 IN NSEC | c.nsec-nonext.com  | A RRSIG NSEC  |

While NSEC provides direct and efficient denial proofs, it exposes domain names in plain-text, enabling zone enumeration. NSEC3 [3] was introduced to mitigate this information leakage by replacing

domain names with cryptographic hashes and ordering them in the hashed namespace. It also introduces additional configuration parameters, including the hash algorithm, salt, and iteration count.

There are two types of negative response:

- **NODATA response:** The queried name exists but has no record of the queried type. NSEC and NSEC3 use a record matching the queried name (or its hash) whose type bitmap shows that the requested record type is absent.
- **NXDOMAIN response:** The queried name does not exist. NSEC typically uses up to two records to prove name and wildcard non-existence, while NSEC3 may use up to three records to additionally prove the closest encloser in the hashed namespace.

**NSEC/3 Variants.** Beyond classic NSEC and NSEC3, deployments exhibit a range of variants with distinct behaviors.

- *Minimally covering NSEC* [40] and *NSEC3 White Lies* [17] dynamically synthesize NSEC/3 records that minimally cover the queried name. For brevity, we collectively refer to these minimally covering techniques as *White Lies* throughout this paper. By generating these bounds on demand, servers reveal little or no information about neighboring zone contents, thereby preventing conventional zone enumeration.
- *Compact Denial of Existence (CDE)* [21] is designed to reduce the overhead of NXDOMAIN responses. Instead of returning multiple NSEC/3 records (up to two for NSEC and three for NSEC3), it encodes non-existence using a single NODATA response that claims the queried name exists but has no data of the requested record type. To preserve the NXDOMAIN signal, RFC 9824 [21] introduces the NXNAME pseudo-type in the NSEC/3 type bitmap.

Examples of NSEC and variants are shown in Figure 1.

### Standardization and Diversity in Variant Response Formats.

The variant mechanisms considered in this paper are documented in RFCs with different publication statuses. *Minimally covering NSEC* [40] and *CDE* [21] are Standards Track, whereas *NSEC3 White Lies* is described in the Informational RFC [17]. These specifications define the required semantics but allow flexibility in the concrete construction of synthesized denial records. Consequently, the same semantic mechanism can appear in different response formats across authoritative-server implementations, motivating our methodology for identifying NSEC/3 variants in Section 6.

**Wildcards and NSEC/3.** DNS wildcards allow a zone to synthesize answers for names that do not explicitly exist; for instance, a wildcard such as “\*.nsec-noext.com” can answer a query for “random.nsec-noext.com” even if that exact name is not present in the zone. This behavior changes the interpretation of negative responses: a query that would otherwise produce NXDOMAIN may instead produce a positive answer or a NODATA response, depending on the queried type and the records associated with the wildcard. Wildcard applicability is governed by the DNS zone tree and can be blocked by a closer existing name, known as the closest encloser. In DNSSEC, denial-of-existence proofs provide authenticated evidence about both wildcard applicability and name non-existence. As a result, comparing responses across hierarchical levels can reveal partial structural information about a zone, which we detail in Section 7.

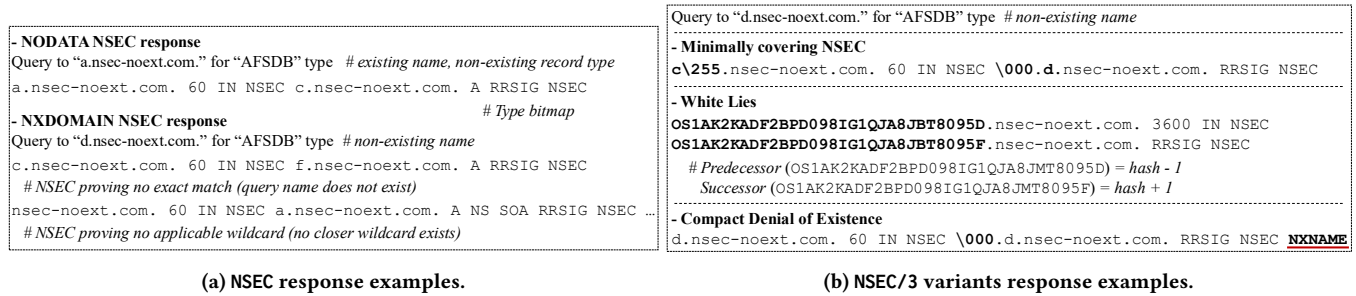


Figure 1: Examples of NSEC and variant responses; RRSIG records are omitted for brevity.

### 3 Related Work

Prior studies have examined authenticated denial of existence in DNSSEC from several perspectives, including its deployment and configuration as well as its operational and security implications.

**Deployment studies.** In 2019, Demke and Deccio [15] studied the size of NSEC3-signed zones and analyzed 2.1 million DNSSEC-signed domains to measure the deployment rate of traditional NSEC/3, NSEC3 *White Lies*, and NSEC “black lies” (later standardized as *CDE*). While their study primarily focused on deployment prevalence, our work expands this scope to include NSEC/3 *CDE* and NSEC *White Lies*. Furthermore, we characterize the structural, naming, provider-level, and wildcard-synthesis behavior of NSEC/3, and identify variants of *White Lies* and *CDE* without assuming a specific synthesized naming pattern, providing a more comprehensive view of DNSSEC negative-response implementations. Daniluk et al. [13] focused on NSEC3 parameter choices and operational configurations. More recently, Huque [20] conducted a brief survey on the use of NSEC/3 variants across Tranco 1M domains, Top-Level Domains (TLDs), and major providers. These studies provide important insights into the deployment of authenticated denial-of-existence mechanisms. Our work complements them with a unified and up-to-date characterization of NSEC/3 variants across diverse domain populations and provider implementations.

**Operational and Security studies.** Other studies have investigated operational and security aspects of authenticated denial of existence beyond deployment measurements. Ashiq et al. [5] analyzed DNSSEC errors observed in DNSViz logs and developed an automated framework for identifying their root causes and suggesting remediation, including errors related to NSEC/NSEC3 proofs. Deccio and Tessem [14] investigated resolver-side support for aggressive negative caching and showed that it is prevalent for NSEC but rarely observed for NSEC3, with substantial variation across resolver implementations. From a security perspective, Gruza et al. [18] demonstrated that computationally expensive NSEC3 proof validation can be exploited to increase resolver CPU load, highlighting the security implications of NSEC3 parameters. These studies focus primarily on DNSSEC operational failures, resolver-side processing, or their security consequences, whereas our work complements them by providing a large-scale characterization of authoritative-side NSEC/NSEC3 deployment and implementation variants, including *White Lies* and *Compact Denial of Existence (CDE)*.

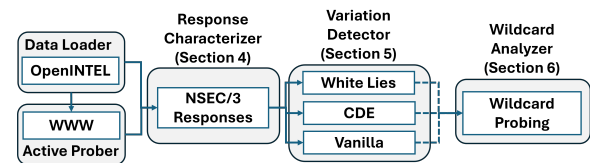


Figure 2: Measurement framework overview. Dotted line suggests potential wild card synthesizing behavior.

**Our work.** Our work differs from prior studies in three ways. First, we provide a more recent and large-scale measurement covering a diverse set of domains and a wider range of NSEC/3 variants. Second, we develop a methodology for identifying NSEC/3 variants across heterogeneous implementations that are difficult to capture with naive approaches (e.g., fixed response pattern matching). Third, we present the first systematic analysis of wildcard synthesis in NSEC/3 deployments. Rather than treating wildcard behavior only as measurement ambiguity, we examine how wildcard synthesis interacts with authenticated denial-of-existence proofs and how this interaction can reveal partial zone structure information. To the best of our knowledge, these aspects have not been systematically explored in prior work.

### 4 NSEC/3 Measurement Framework

We develop a measurement framework to characterize NSEC/3 deployment in the wild. As illustrated in Figure 2, the framework integrates large-scale DNS data with active probing and a unified analysis pipeline. Our main goal is to accurately identify NSEC/3 variants under real-world complexities and characterize how wildcard synthesis interacts with authenticated denial-of-existence proofs.

**Data Sources.** We use DNS data from OpenINTEL [30, 39] as our primary data source. The platform scans a wide range of DNS records, including SOA, NS, DNSKEY, DS, RRSIG, and NSEC/3-related records. Our framework leverages two types of datasets:

- **.com, .net, .org zone data:** These datasets cover all apex domains under three Top-level domains (TLDs). Throughout the paper, we refer to these domains as *apex* domains.
- **Four popular rankings:** These datasets cover domains from four popularity-based lists—Cisco Umbrella 1M [10], Tranco 1M [32], Cloudflare Radar 1M [12], and Majestic 1M [25]. Because

|          | Source domains (from OpenINTEL) |           |       |           |        |           |        | WWW subdomains (from our active prober) |           |        |           |        |
|----------|---------------------------------|-----------|-------|-----------|--------|-----------|--------|-----------------------------------------|-----------|--------|-----------|--------|
|          | SOA                             | w/ DNSKEY |       | NSEC      |        | NSEC3     |        | w/ DNSKEY                               | NSEC      |        | NSEC3     |        |
|          | #. Apex                         | #. Apex   | %     | #. Apex   | %      | #. Apex   | %      | #                                       | #         | %      | #         | %      |
| .com     | 149,021,866                     | 8,364,107 | 5.61% | 1,577,074 | 18.86% | 6,720,037 | 80.34% | 8,364,107                               | 1,463,427 | 17.50% | 6,561,180 | 78.44% |
| .net     | 11,156,231                      | 707,851   | 6.34% | 160,534   | 22.68% | 543,279   | 76.75% | 707,851                                 | 155,078   | 21.91% | 532,883   | 75.28% |
| .org     | 10,972,417                      | 707,896   | 6.45% | 146,299   | 20.67% | 556,602   | 78.63% | 707,896                                 | 140,760   | 19.88% | 548,415   | 77.47% |
|          | #. FQDN                         | #. FQDN   | %     | #. FQDN   | %      | #. FQDN   | %      | #                                       | #         | %      | #         | %      |
| Umbrella | 384,922                         | 26,034    | 6.76% | 16,251    | 62.42% | 9,523     | 36.58% | 20,731                                  | 13,004    | 62.73% | 6,952     | 33.53% |
| Tranco   | 968,036                         | 79,790    | 8.24% | 49,080    | 61.51% | 30,109    | 37.74% | 71,512                                  | 42,812    | 59.87% | 27,054    | 37.83% |
| Radar    | 1,137,219                       | 102,067   | 8.98% | 65,042    | 63.72% | 36,246    | 35.51% | 93,880                                  | 58,725    | 62.55% | 33,327    | 35.50% |
| Majestic | 946,890                         | 75,863    | 8.01% | 40,987    | 54.03% | 34,296    | 45.21% | 69,995                                  | 37,372    | 53.39% | 31,237    | 44.63% |

**Table 1: DNSSEC and NSEC/3 adoption rate as of April 6, 2026. For source domains, TLD zone datasets are measured at the apex-domain level, whereas popularity ranking datasets are measured at the FQDN level. The NSEC/3 percentages are calculated relative to the total number of domains with a DNSKEY record.**

these lists may contain both apex and subdomains, we refer to them collectively as *FQDN* (Fully Qualified Domain Names).

**NSEC/3 probing.** To elicit denial-of-existence responses, OpenINTEL prepends a likely non-existent label (e.g., “123-nonexistent-dnsjedi-456”) to a target domain (i.e., apex or FQDN) and issues a DNS query for the AFSDB record type. It does not use A records, which should theoretically trigger negative responses given the non-existent label, but A records are more likely to be affected by wildcard synthesis or provider-specific default responses, which can mask denial-of-existence behavior. On the other hand, using non-existent or customized record types may be subject to filtering or inconsistent handling by resolvers and authoritative servers, leading to unreliable measurements. Therefore, AFSDB is used because it is a valid but deprecated record type [2] that is rarely deployed in practice, making it suitable for triggering negative responses.

**Auxiliary DNSKEY scan.** In the OpenINTEL datasets, we observe that 5–7% of domains publish DNSKEY records but do not have corresponding NSEC or NSEC3 records. To avoid treating these domains as lacking NSEC/3 solely based on a single data source, we perform an auxiliary scan using the same probing methodology described above.<sup>1</sup> Among these domains, we find that 90% return NSEC/3 records in our auxiliary scan. We therefore use the auxiliary scan to supplement the OpenINTEL data: domains for which we observe NSEC or NSEC3 in the auxiliary scan are treated as NSEC/3-enabled and included in the corresponding datasets for the subsequent analysis.

## 4.1 Framework Details

Our framework consists of three main modules: (i) data loader, (ii) active prober, and (iii) NSEC/3 analyzer.

**4.1.1 Data Loader.** This module ingests DNS data from OpenINTEL. For TLD zone datasets, we access the data through an agreement with OpenINTEL and retrieve it from their storage. For popular ranking datasets, which are publicly available, we fetch them

directly from the OpenINTEL websites [29]. The module preprocesses the retrieved data (e.g., identifying DNSSEC-enabled domains and generating the apex domain list) and stores the processed data for subsequent analysis.

**4.1.2 Active Prober.** The active prober performs additional measurements on WWW subdomains. We focus on WWW because it is widely used in practice and often represents the web-facing entry point of a domain, making it an important target for understanding real-world deployment behavior. Additionally, WWW subdomains may be delegated to authoritative nameservers different from those of their corresponding apex domains, resulting in potentially different NSEC/3 configurations. This allows us to capture deployment discrepancies across hierarchical levels. Furthermore, the WWW probing results serve as an input to our wildcard-aware analysis.

The active prober targets DNSSEC-enabled apex domains produced by the data loader. For FQDNs from popularity ranking datasets, entries may include subdomains rather than apex domains; in such cases, we extract the corresponding apex (e.g., “nsec-noext.com”<sup>2</sup> from “sub.nsec-noext.com”). To ensure consistency with the OpenINTEL data, the active prober adopts the same probing strategy: issuing AFSDB queries for non-existent labels under the www hierarchy (e.g., “123-nonexistent-dnsquery.www.{apex domain}”). The active prober uses Google Public DNS resolver (8.8.8.8) as its default resolver for scanning.

**4.1.3 NSEC/3 Analyzer.** It comprises three sub-modules:

- A **response characterizer**, which computes high-level ecosystem statistics, such as NSEC/3 adoption, record structures, and authoritative name server prevalence.
- An **NSEC/3 variation detector**, which applies novel heuristic-based methodology to distinguish NSEC/3 variants (e.g., *White Lies* and *CDE*) from vanilla NSEC/3, while capturing implementation-specific patterns across various DNS providers.
- A **wildcard analyzer**, which identifies wildcard-enabled zones and characterizes how wildcard behavior changes across different levels of the domain hierarchy. This allows us to examine how wildcard behavior interacts with denial-of-existence proofs and what partial zone structure information can be inferred.

<sup>1</sup>These cases may arise from several factors, such as measurement timing or incomplete DNSSEC deployments. We are currently discussing this case with the OpenINTEL team.

<sup>2</sup>A non-existent domain name used for anonymization.

## 4.2 Measurement Period

With our framework, we present our measurement results using the OpenINTEL snapshot on April 6, 2026 along with subsequent additional scans that immediately followed. The snapshot covers 174 million domains across three TLDs and four popular domain rankings.

## 5 NSEC/3 Deployment in the Wild

We first analyze NSEC/3 adoption rate, response structure, and parameter configuration. We examine deployment from two perspectives: (1) *Source domains*, which include all domains in the OpenINTEL source datasets and may contain both apex and FQDNs, as described in Section 4; and (2) *WWW subdomains*, which corresponds to the datasets generated by our active prober (Section 4.1.2).

**DNSSEC Adoption.** We first restrict our analysis to domains for which we successfully observed an SOA record at the zone apex, establishing a baseline of active and resolvable domains. Among these baseline domains, we determine DNSSEC adoption based on the presence of DNSKEY records. As shown in Table 1, DNSSEC support is around 6% for domains in .com, .net, and .org, and ranges from 6% to 9% for domains in the popularity rankings. Because our WWW subdomain scan targets only apex domains with DNSKEY (Section 4.1.2), we do not separately report DNSSEC adoption rates for WWW subdomains. We focus on these DNSSEC-enabled domains in all subsequent analyses.

### 5.1 NSEC/3 Adoption

We first examine how widely NSEC/3 is deployed in practice, with the results shown in Table 1. For source domains, we observe a clear divergence in deployment practices: domains in the three TLD datasets show a trend opposite to those in the popularity ranking datasets. In the .com, .net, and .org zones, NSEC3 is dominant, accounting for more than 75% of DNSSEC-enabled domains. In contrast, NSEC is dominant (i.e., nearly 60%) in the popularity rankings. This difference suggests that NSEC/3 deployment practices vary across domain populations. Domains in TLD zone datasets are dominated by NSEC3, whereas domains appearing in popularity rankings more commonly use NSEC. For WWW subdomains, the overall trends remain similar to those observed for the source domains, with only small changes in NSEC/3 deployment rates (by < 1% on average across datasets).

**Adoption trend evolving since 2019.** Compared to prior study in 2019 [15], we observe a substantial increase in the NSEC/3 adoption, from 112K to 1.5M domains for NSEC and from 725K to 6.7M domains for NSEC3 in .com, indicating more than an order-of-magnitude increase in deployment over the past several years.

**Domains with DNSKEY but without NSEC/3.** We observe that approximately 1% of domains across our datasets publish DNSKEY records but do not return corresponding NSEC/3 responses. We further investigate these cases through active probing and confirm that these domains do not return positive answers (i.e., no AFSDb records are observed for high-entropy queried names), suggesting that denial-of-existence responses should have been triggered. We identify several underlying causes, which are detailed in Appendix B.

|          | NSEC   |        | NSEC3  |        |
|----------|--------|--------|--------|--------|
|          | Source | WWW    | Source | WWW    |
| .com     | 76.57% | 78.16% | 90.15% | 80.90% |
| .net     | 80.74% | 81.32% | 90.44% | 84.50% |
| .org     | 81.38% | 81.98% | 91.11% | 81.73% |
| Umbrella | 69.40% | 68.57% | 76.80% | 75.45% |
| Tranco   | 62.90% | 63.16% | 80.03% | 77.52% |
| Radar    | 62.88% | 63.50% | 82.03% | 80.38% |
| Majestic | 65.12% | 66.14% | 82.52% | 80.46% |

Table 2: Proportion of validated responses.

**Domain with both NSEC and NSEC3** We observed 1,443, and 441 domains from source and WWW domains containing both NSEC and NSEC3 records within a single resolution chain (e.g., across CNAME redirects). This typically stems from the CNAME delegation where the original zone and the delegated zone deploy different authenticated denial mechanism. These "compound" responses highlight the complexity of validating negative results in heterogeneous environments.

### 5.2 NSEC/3 Response Characteristics

We characterize NSEC/3 responses, focusing on authentication rate, the structure of returned resource records, and anomalies observed in denial-of-existence responses.

**DNSSEC-validated responses.** To examine whether NSEC/3 responses are successfully validated by the recursive resolver, we inspect the AD (Authenticated Data) flag [34]. The AD flag indicates that the resolver has cryptographically validated the answer using DNSSEC. For each dataset, we compute the proportion of responses with the AD set, as summarized in Table 2. Overall, approximately 63–81% of NSEC responses are validated, compared with 77–91% for NSEC3. Domains from three TLD datasets exhibit slightly higher authentication rates than domains from popularity-based lists; by roughly 20% for NSEC and 5–10% for NSEC3. Additionally, domains using NSEC3 show consistently higher authentication rates than those using NSEC (approximately 15% on average).

A missing AD in responses does not necessarily indicate invalid or corrupt DNSSEC data; it may instead reflect that the resolver cannot establish a full secure delegation chain to the zone. For example, this can occur when the child zone lacks a corresponding DS record in the parent [9] or when a CNAME chain traverses unsigned or insecure target domains.

**Response structure.** We examine the number of Resource Records (RRs) returned per query, as summarized in Table 3. In this analysis, we only count NSEC/3 RRs, excluding accompanying records such as RRSIG or other auxiliary resource records. We observe that for both TLD and popularity-based datasets, the majority of NSEC responses contain a single RR per query. For NSEC3, over 44% of domains from TLD datasets return two RRs per query, while more than 74% of popularity ranking-based domains return three RRs; this higher RR count is expected as NSEC3 typically requires multiple records to construct a "closest encloser" proof, demonstrating both the existence of the nearest ancestor and the non-existence of the queried name [3]. The distribution for WWW subdomains largely

| #. RRs   | Source domains (from OpenINTEL) |       |       |        |        |        |       | WWW subdomains (from our active prober) |        |       |        |        |        |       |
|----------|---------------------------------|-------|-------|--------|--------|--------|-------|-----------------------------------------|--------|-------|--------|--------|--------|-------|
|          | NSEC                            |       |       | NSEC3  |        |        |       | NSEC                                    |        |       | NSEC3  |        |        |       |
|          | 1                               | 2     | 3     | 1      | 2      | 3      | 4     | 1                                       | 2      | 3     | 1      | 2      | 3      | 4     |
| .com     | 97.85%                          | 2.14% | 0.01% | 12.05% | 44.61% | 43.34% | 0.00% | 89.64%                                  | 10.35% | 0.01% | 11.94% | 44.00% | 44.06% | 0.00% |
| .net     | 96.66%                          | 3.34% | 0.01% | 12.93% | 46.01% | 41.05% | 0.00% | 89.56%                                  | 10.44% | 0.00% | 12.78% | 45.48% | 41.74% | 0.00% |
| .org     | 97.06%                          | 2.94% | 0.00% | 11.72% | 44.72% | 43.56% | 0.00% | 90.67%                                  | 9.33%  | 0.00% | 11.57% | 44.26% | 44.16% | 0.00% |
| Umbrella | 92.83%                          | 7.17% | 0.00% | 4.91%  | 15.84% | 79.25% | 0.00% | 94.33%                                  | 5.67%  | 0.00% | 4.09%  | 14.86% | 81.06% | 0.00% |
| Tranco   | 96.62%                          | 3.38% | 0.00% | 4.06%  | 17.31% | 78.63% | 0.01% | 95.90%                                  | 4.10%  | 0.00% | 3.40%  | 16.99% | 79.60% | 0.00% |
| Radar    | 97.52%                          | 2.48% | 0.00% | 5.04%  | 18.63% | 76.32% | 0.01% | 96.48%                                  | 3.52%  | 0.00% | 4.21%  | 17.98% | 77.81% | 0.00% |
| Majestic | 96.74%                          | 3.25% | 0.01% | 5.25%  | 19.77% | 74.97% | 0.01% | 96.71%                                  | 3.29%  | 0.01% | 4.57%  | 19.38% | 76.05% | 0.00% |

**Table 3: Count of NSEC/3 resource records (RRs) per requests. Note that we only count NSEC/3 RRs, excluding accompanying records such as RRSIG or other auxiliary resource records.**

|          | Source |        | WWW    |        |
|----------|--------|--------|--------|--------|
|          | NSEC   | NSEC3  | NSEC   | NSEC3  |
| .com     | 41.50% | 89.11% | 52.43% | 94.56% |
| .net     | 31.77% | 88.32% | 37.78% | 93.43% |
| .org     | 35.03% | 89.42% | 41.63% | 94.39% |
| Umbrella | 19.41% | 82.34% | 18.48% | 92.13% |
| Tranco   | 21.25% | 77.14% | 21.52% | 90.57% |
| Radar    | 16.46% | 73.48% | 18.11% | 88.37% |
| Majestic | 21.39% | 77.14% | 25.34% | 92.10% |

**Table 4: Proportion of NXDOMAIN responses. The remaining percentage (100% – NXDOMAIN) represents NODATA responses.**

mirrors that of apex domains. However, one exception is that, in the TLD zone datasets, the proportion of single-RR NSEC responses is about 7% lower for WWW subdomains than for source domains. This suggests that denial-of-existence behavior can vary across different levels of the domain hierarchy.

**Extra responses due to CNAME.** Authenticated denial-of-existence proofs typically contain at most two NSEC or three NSEC3 records [17]. However, we identify 182 and 243 responses containing three NSEC and four NSEC3, respectively (Table 3). A closer examination reveals that 99.99% of these instances stem from CNAME redirections. In such cases, the response can include denial-of-existence proofs for both the original query and the target name pointed to by the CNAME record; for instance, if a query triggers a wildcard match that resolves to a CNAME, the response may include both a *wildcard expansion proof* from the source zone and a NODATA proof from the target zone.

**NXDOMAIN and NODATA Responses.** Table 4 reports the proportion of NXDOMAIN responses across our datasets; the corresponding NODATA proportion is complementary. We observe that queries to WWW subdomains yield a significantly higher proportion of NXDOMAIN responses compared to queries to source domains. This difference is likely related to the potential wildcard records at the apex level, which can convert what would otherwise be an NXDOMAIN response into a NODATA response. The higher NXDOMAIN ratio observed for WWW queries suggests that wildcard synthesis is less commonly applied at deeper subdomain levels. We provide a detailed analysis of wildcard behavior and its interaction with NSEC/3 mechanisms in Section 7.

### 5.3 Name Server Providers

The response level differences observed above motivate a closer look at the authoritative name server providers behind these domains. We therefore characterize the role of authoritative name server providers in shaping the denial-of-existence behavior.

To identify authoritative name server providers, we extract NS records for each domain from the data sources (i.e., OpenINTEL datasets). For our WWW measurements, we additionally issue NS queries to the subdomains to obtain their corresponding name server information. When no NS record is available, we leverage the SOA MNAME field in the authority section of the negative response as an indicator of the authoritative name servers. We group name servers based on their Second-Level Domain (SLD). For providers that encode identifiers using digits in the SLD (e.g., awsdns-XX), we normalize them using regular expressions.

**NSEC record.** We observe that Cloudflare is the most common provider across all NSEC deployments. In the TLD zone datasets, it is followed by Infomaniak and GoDaddy (via registrar-servers). In the popularity-based datasets, Cloudflare remains dominant (over 60%), followed by AWS and Domain Control.

**NSEC3 record.** For NSEC3 deployments, Google, wix.com, and OVH Cloud are the primary providers in TLD-zone datasets. In popularity-based datasets, Google’s share is more limited (approximately 10–20%), and no single provider dominates to the same extent as Cloudflare does in NSEC deployments. The top five provider distributions are provided in Table 13 (in Section C).

We further compare provider distributions between source domain and WWW-subdomain measurements. Overall, the dominant providers for WWW subdomains closely mirror those observed for the source domains. Meanwhile, although the distribution of response structures and NXDOMAIN ratios differs between apex and subdomain queries, e.g., approximately 97% of apex NSEC responses contain a single resource record compared to about 90% for WWW queries, we do not observe significant shifts in the set of dominant providers. This suggests that the observed structural differences are primarily driven by query depth rather than changes in authoritative name server providers.

## 6 NSEC/3 Variant in the Wild

We next examine NSEC/3 variations, focusing on identifying *White Lies* and Compact Denial of Existence (CDE). Distinguishing these

from vanilla deployments is challenging due to diverse implementation strategies across providers.

**Challenges in identifying *White Lies*.** *White Lies* rely on synthesizing minimally covering domain names to prevent zone enumeration. For NSEC3, RFC7129 [17] suggests an epsilon construction where the predecessor and successor hashes are derived by decrementing or incrementing the queried hash value. Due to its computational efficiency, this  $\pm 1$  offset method is widely adopted, making *White Lies* in NSEC3 relatively consistent and detectable.

In contrast, NSEC implementations exhibit greater diversity. While RFC4470 [40] introduces an epsilon function for NSEC, it is not strictly mandated. Due to the excessive domain name length introduced by padding the leftmost label with zeros [40], providers commonly adopt custom synthesis strategies. Therefore, the detection of NSEC *White Lies* is non-trivial and its variability necessitates a systematic measurement of real-world synthesis strategies.

**Challenges in identifying *CDE*.** Identifying *CDE* is challenging because it is structurally similar to *White Lies*: both use minimally covering records to provide authenticated denial of existence. However, unlike *White Lies*, which preserve the distinction between NXDOMAIN and NODATA, *CDE* collapses both outcomes into a single NODATA response. The ambiguity further compounded by wildcard synthesis, which can also yield NODATA for non-existent names.

While the NXNAME pseudo-type (RFC 9824 [21]) provides a strong indicator of *CDE*, its adoption among authoritative name server providers is not universal. Without NXNAME, it is difficult to distinguish *CDE*-generated NODATA responses from those produced by *White Lies*.

Consequently, reliably identifying *CDE* requires analysis beyond individual response inspection. In particular, it necessitates aggregated behavioral signals, such as the near absence of NXDOMAIN responses across large query sets and the prevalence of single resource record (RR) answers.

To address these challenges, we systematically measure NSEC/3 variations in real-world deployments. Our analysis is guided by the following research questions: (i) what minimally covering algorithms are being deployed? (ii) which name server providers implement specific NSEC/3 variations? (iii) what are the adoption rates of these variations? and (iv) what is the current state of NXNAME adoption?

## 6.1 Methodology

We develop a pipeline to systematically identify NSEC/3 *White Lies* and *CDE* in real-world deployments. To the best of our knowledge, this is the first systematic approach to jointly detect both variations across NSEC and NSEC3.

Prior work [15] focused on identifying NSEC3 *White Lies* using  $\pm 1$  hash offsets and detecting NSEC *CDE* via simple pattern matching (e.g., comparing “foo.com” to “\000.foo.com”). In contrast, our approach leverages structural properties of responses, enabling detection across a broader range of implementation-specific behaviors.

Our framework utilizes a two-stage approach:

- **Filtering minimally covering responses:** Both *White Lies* and *CDE* employ minimally covering mechanisms to mitigate zone enumeration. As a result, the lexicographical distance between

the query name and its predecessor and successor is significantly smaller than in vanilla implementations. We introduce a *DNS Canonical Distance* metric (Section 6.2) to quantify this property and identify minimally covering candidates.

- **Identifying *White Lies* and *CDE* deployments:** We then categorize name server providers based on their aggregated minimally covering response behavior. Providers deploying *CDE* are expected to exhibit a dominant proportion of NODATA responses, whereas *White Lies* deployments retain a mix of NODATA and NXDOMAIN.

Finally, we correlate our analysis from both stages and identify unique behavioral “fingerprints” for different NSEC/3 implementations. This enables us to generalize our findings and providing a robust estimate of *CDE* and *White Lies* adoption rates across the global NSEC/3 ecosystem.

## 6.2 DNS Canonical Distance

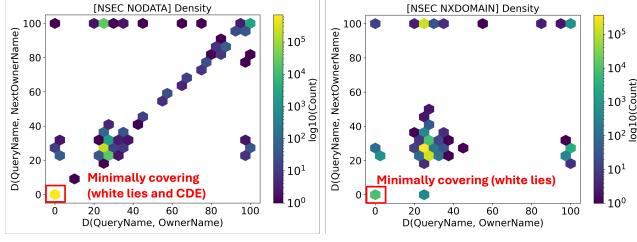
We define a *DNS Canonical Distance* metric to identify minimally covering responses. The metric captures the structural proximity between a queried domain name and the owner (and next owner) names returned in NSEC/3 responses. Since NSEC operates over plaintext domain names while NSEC3 uses hashed representations, we define two corresponding formulations.

**6.2.1 NSEC distances.** We define a hierarchical distance metric to measure the proximity between an NSEC owner (or next owner) name and a query name. Let  $\hat{Q}$  and  $\hat{N}$  denote the normalized forms of the query name  $Q$  and the NSEC name  $N$ , respectively. We use regular expression to normalize the string by removing escaped octet sequences (e.g., \255, \000) that typically used in minimal covering name synthesis and retaining only letters, digits, and hyphens (legitimate second-level domain characters).

We then compute the Levenshtein distance  $\text{Lev}(s_1, s_2)$  between corresponding labels, proceeding from right to left (beginning with the top-level domain). The Levenshtein distance represents the minimum number of single-character edits required to transform one string into another [27]. To reflect the DNS hierarchy, the distance at depth  $i$  is weighted by a factor of  $10^i$ . If the domain names contain a different number of labels, missing labels are treated as empty strings. To avoid extreme values arising from highly dissimilar long domain names, we apply a saturation threshold  $T = 100$ . The threshold is used primarily for computational efficiency, as our analysis focuses on distances near zero, setting  $T = 100$  or any sufficiently large value does not materially affect our results, as long as the threshold is not chosen close to zero. Thus, the final distance is defined as:

$$D(\hat{Q}, \hat{N}) = \min \left( T, \sum_{i=0}^{\max(L_{\hat{Q}}, L_{\hat{N}})-1} 10^i \cdot \text{Lev}(\hat{q}_i, \hat{n}_i) \right) \quad (1)$$

where  $\hat{q}_i$  and  $\hat{n}_i$  denote the  $i$ -th labels from the right in  $\hat{Q}$  and  $\hat{N}$ , respectively, with  $i = 0$  corresponding to the top-level domain. A low distance suggests that the two names are structurally and lexicographically similar. and the clamping ensures that our analysis focused on the minimally covering activity while treating all significantly distant records as a uniform “vanilla” population.



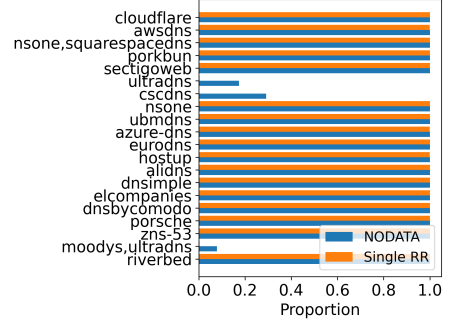
**Figure 3: Joint density of NSEC owner and next owner distances in .com zone. The cluster near the origin (0, 0) indicates minimally covering records.**

**NSEC distances in the wild.** Figure 3 presents a hexbin density plot correlating the DNS Canonical Distances of the query name to the NSEC owner name (x-axis) and next owner name (y-axis) in NODATA and NXDOMAIN responses respectively. The color scale represents the logarithmic frequency ( $\log_{10}$ ) of occurrences. We observe a concentrated cluster around the origin (0, 0) where  $D \leq 1$ , representing the minimally covering variations (*White Lies* and *CDE*). More specifically, 73.44% of NODATA and 1.5% of NXDOMAIN resource records in the .com zone exhibit a distance  $D \leq 1$ . In these instances, both the owner and next owner names are nearly identical to the query name after normalization, resulting in near-zero distances. Conversely, vanilla NSEC deployments have much higher distance values (typically above 20). The significant spatial gap between the (0, 0) cluster and the vanilla distribution demonstrates that our distance metric provides a clear decision boundary for programmatically isolating *White Lies* and *CDE* from the standard version. We observe similar patterns across all datasets and for WWW subdomains.

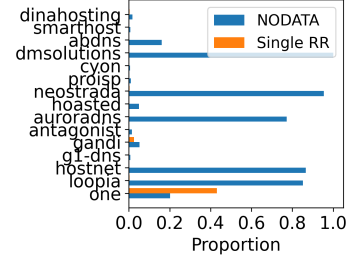
**6.2.2 NSEC3 distance.** Since NSEC3 represents owner and next owner names as hashed values, we measure their proximity by interpreting the hashes as large integers. The distance is then defined as the absolute numerical interval between the hashed predecessor and successor.

**NSEC3 distances in the wild.** Figure 6 in Appendix D illustrates the NSEC3 distances on a logarithmic scale, revealing a disparity consistent with our plaintext NSEC observations. We find that 33.97% of NODATA records and 5.96% of NXDOMAIN records in the .com zone exhibit a distance of  $D \leq 2$ . In contrast, the remainder of the population exhibits significantly larger distances. This clear separation in the log-scale distribution further supports the use of a distance-based threshold to distinguish hashed minimally covering from standard NSEC3 deployments.

**Identical NSEC3 hashes.** We observe 47,424 cases where the NSEC3 owner and next-owner hashes are identical in .com zone. This occurs when the zone contains only a single signed name (the apex), causing the NSEC3 record to point back to itself. As a result, all non-existent queries in such zones return the same NSEC3 record, corresponding to the hash of the zone apex itself. Although these records exhibit a minimal hash distance, we exclude them from our analysis as they represent a structural artifact of small zones rather than an intentional minimally covering mechanism.



**(a) NSEC top 20 name servers.**



**(b) NSEC3 top 15 name servers.**

**Figure 4: Per-name server distribution of NODATA proportion and the proportion of responses returning a single resource record per query in .com zone.**

**Takeaways.** The canonical and hash-based distance metrics serves as a robust metric for programmatically distinguishing minimally covering variations, such as *White Lies* and *CDE*, from standard NSEC/3 deployments.

### 6.3 Name Server Categorization

After filtering out minimally covering responses, we then categorize their corresponding name service providers from two criteria:

- **Proportion of NODATA responses:** A key distinguishing factor between *White Lies* and *CDE* is that the latter exclusively generates NODATA responses, regardless of whether the queried name actually exists. Consequently, providers deploying *CDE* exhibit a significant skew toward NODATA responses, whereas those utilizing *White Lies* produce a mix of NODATA and NXDOMAIN responses.
- **Proportion of single-RR responses:** Another distinguishing characteristic is that *CDE* typically returns only one resource record per query. As a result, name servers deploying *CDE* exhibit a high proportion of queries returning exactly one RR.

We use .com zone data to demonstrate our methodology here and the resulting distributions, which highlight distinct behavioral profiles across providers, are shown in Figure 4.

**6.3.1 Prevalence of CDE in NSEC.** Figure 4a shows the proportions of NODATA responses and single-RR responses for the top 20 name servers observed in NSEC records in the .com zone. We observe clear disparity across providers. Most name servers (e.g.,

| Pattern | Observed Providers                         | Owner Name                                                                                       | Next Owner Name                                                     | Functionality                                       | Variation         |
|---------|--------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------|-------------------|
| 1       | Cloudflare, AWS, NS1, Azure, EuroDNS, etc. | nonexistent.nsec-noext.com                                                                       | \000.nonexistent.nsec-noext.com                                     | Deny QNAME                                          | <i>CDE</i>        |
| 2       | UltraDNS, CSCDNS*, DomainNetwork*, etc.    | {{(-)^{*}}}.{nonexistent-1}~.nsec-noext.com<br>{{(-)^{*}}}.!~.nsec-noext.com<br>*.nsec-noext.com | nonexistent!.nsec-noext.com<br>~.nsec-noext.com<br>~.nsec-noext.com | Deny QNAME<br>Deny Wildcard<br>Acknowledge Wildcard | <i>White Lies</i> |
| 3       | w3h.io                                     | {nonexistent-1}\255.nsec-noext.com<br>\255.nsec-noext.com                                        | nonexistent\000.nsec-noext.com<br>*\000.nsec-noext.com              | Deny QNAME<br>Deny Wildcard                         | <i>White Lies</i> |

\* Indicates providers that potentially use UltraDNS backend service.

**Table 5: Unique patterns observed for minimally covering NSEC responses (query: nonexistent.nsec-noext.com). We use {nonexistent-1} to represent the decrement of the final character in the query label, and use (-)^{\*} to suggest that the “~” can repeat 0 or more times depending on the depths of the domain name.**

|          | Source domains (from OpenINTEL) |            |                   |            | WWW subdomains (from our active prober) |            |                   |            |
|----------|---------------------------------|------------|-------------------|------------|-----------------------------------------|------------|-------------------|------------|
|          | NSEC                            |            | NSEC3             |            | NSEC                                    |            | NSEC              |            |
|          | <i>White Lies</i>               | <i>CDE</i> | <i>White Lies</i> | <i>CDE</i> | <i>White Lies</i>                       | <i>CDE</i> | <i>White Lies</i> | <i>CDE</i> |
| .com     | 0.74%                           | 43.96%     | 6.98%             | 0.00%      | 0.78%                                   | 41.92%     | 16.88%            | 0.00%      |
| .net     | 0.96%                           | 56.53%     | 6.86%             | 0.00%      | 0.98%                                   | 57.47%     | 13.21%            | 0.00%      |
| .org     | 0.80%                           | 52.75%     | 5.81%             | 0.00%      | 0.83%                                   | 53.75%     | 15.91%            | 0.00%      |
| Umbrella | 1.71%                           | 76.40%     | 8.11%             | 0.01%      | 1.84%                                   | 80.14%     | 10.11%            | 0.00%      |
| Tranco   | 1.31%                           | 74.84%     | 9.31%             | 0.00%      | 1.37%                                   | 77.16%     | 13.90%            | 0.00%      |
| Radar    | 0.99%                           | 79.17%     | 10.37%            | 0.00%      | 0.96%                                   | 80.32%     | 14.31%            | 0.00%      |
| Majestic | 1.55%                           | 73.04%     | 11.19%            | 0.00%      | 1.52%                                   | 73.23%     | 15.82%            | 0.00%      |

**Table 6: *White Lies* and *CDE* adoption rate.**

Cloudflare, AWS DNS, NS1, and Azure DNS) exhibit proportions close to 100% for both metrics, indicating that nearly all queries return NODATA responses with a single RR, consistent with compact denial of existence (*CDE*) behavior.

In contrast, a small subset of providers, most notably UltraDNS and CSCDNS, show significantly lower proportions, returning a mix of NODATA and NXDOMAIN responses and typically return multiple RRs per query, suggesting the deployment of *White Lies*.

Overall, the figure reveals a sharp separation between providers deploying *CDE* and those using *White Lies*, with *CDE* dominating among most large-scale operators. Our observations align with public technical documentation from Cloudflare [11], AWS Route 53 [6], and Azure [26], as well as recent industry surveys [20]. Furthermore, our methodology successfully identifies several additional providers displaying these behaviors.

**6.3.2 Absence of *CDE* in NSEC3.** In contrast to NSEC, where *CDE* is widely deployed, we find no evidence of *CDE* usage within NSEC3 deployments, as illustrated in Figure 4b. Although one specific name server (“dmsolutions.se”) exhibits a 100% NODATA ratio, its responses consistently contain multiple resource records, which is characteristic of standard NSEC3 non-existence proofs rather than the minimized structure of *CDE*. Overall, our findings indicate that all dominant name server providers utilizing minimally covering techniques in NSEC3 favor *White Lies* over *CDE*.

**Takeaways.** Combining the ratio of NODATA responses with single-RR response counts cleanly separates *CDE* and *White Lies* deployments. Using these signals, we find that *CDE* dominates among major NSEC operators, whereas NSEC3 deployments predominantly rely on *White Lies*.

## 6.4 Characterizing *White Lies* and *CDE*

We conduct a thorough analysis across all datasets using the above metrics to study the minimally covering strategy, the adoption ratio and current state of NXNAME adoption in *CDE*.

**6.4.1 Minimally covering strategy.** We identify three distinct strategies used by providers to generate minimally covering NSEC records, as summarized in Table 5. These patterns were derived through an iterative manual examination of the minimally covering responses: we first identified a recurring structural pattern, filtered the observations matching that pattern, and then examined the remaining responses to identify additional patterns. This process continued until no substantial new patterns emerged.

The most prevalent approach, utilized by major providers such as Cloudflare and AWS, employs a null-byte prefix (**Pattern 1**). Other providers utilize character decrementing and symbolic wrapping techniques (**Patterns 2 and 3**) to create the smallest possible lexicographical gap around the nonexistent query. These techniques are deployed both to *deny* the existence of a wildcard and, in specific cases, to *prove* the existence of a wildcard that covers the query. Notably, for character-decrementing techniques, the specific character alphabet may vary between implementations, for instance,

UltraDNS includes specific characters in its shifting logic and its “~” can repeat zero or more times based on the depth of the domain name. **Pattern 3** can be viewed as a variation of the epsilon function proposed in RFC4470 [40], where the predecessor is padded with a single “\255”. This design reduces the amount of bytes transmitted compared to the original construction.

We further examine whether specific minimally covering patterns are tied to *CDE* or *White Lies* implementations. Our analysis reveals that **Pattern 1** (Null-byte prefix) is exclusively utilized in NSEC *CDE* deployments. Notably, these deployments yield zero NXDOMAIN responses across all datasets. In contrast, the remaining two patterns are exclusively associated with NSEC white lie implementations.

**Validation.** We cross-reference our observations with vendor documentation and other empirical studies to validate our observations. Public technical documentation from Cloudflare [11], AWS [6], and Azure [26] confirms our findings in Table 5, while a reverse-engineering study of UltraDNS [37] reports behavior similar to our observations. For patterns without external confirmation, we have initiated outreach to the relevant providers and plan to seek direct confirmation of our findings.

**6.4.2 Adoption of White Lies and CDE.** By applying the patterns defined in Table 5, we compute the adoption rates of *White Lies* and *CDE* across our NSEC/3 dataset, with results summarized in Table 6. We observe that *CDE* dominates over *White Lies* in NSEC deployments, whereas *CDE* is rarely observed in NSEC3 deployments. Overall, the proportion of minimally covering techniques exceeds 40% in NSEC but is only around 7% in NSEC3, indicating a significantly lower adoption in the latter. One possible explanation is that NSEC3 inherently mitigates zone enumeration through hashed owner names, reducing the need for additional obfuscation mechanisms such as *CDE* and *White Lies*.

We further observe a higher *CDE* adoption rate among domains in popularity lists compared to zone file datasets, likely driven by the dominance of providers such as Cloudflare in popular domains (Table 13).

Notably, while there is no evidence of major name server providers adopting NSEC3 *CDE* on a large scale, we identified two specific domains ({nonexistent}.delta.mil.gov.ua from the Umbrella list and {nonexistent}.www.sslforshaz.com from the WWW scan) utilizing Cloudflare as their provider that appear to use NSEC3 *CDE*. These two domains were identified through the presence of the NXNAME pseudo-type, a behavior we examine next. While Cloudflare predominantly deploys NSEC for *CDE*, these observations suggest that the mechanism is compatible with NSEC3 and can be deployed under this setting.

**6.4.3 Current deployment status of NXNAME.** The NXNAME pseudo-type was standardized in RFC 9824 [21] in 2025, and its real-world adoption has received limited study to date. We examine the extent to which authoritative name server providers have adopted NXNAME.

Focusing on the top 15 name server providers identified as deploying *CDE* across all datasets, we analyze whether they also support NXNAME by examining the NSEC Type Bit Map. Table 7 summarizes our findings. We observe that major providers such as Cloudflare and NS1 have deployed NXNAME, whereas others, including AWS Route53 and Azure DNS, have not yet adopted this newly

| Deployed                                                                                                | Not yet deployed                                           |
|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Cloudflare, NS1, Porkbun*, SectigoWeb, HostUp*, elcompanies.com*, DNSimple, ubmdns.com*, FoundationDNS* | Route53, AzureDNS, EudoroDNS, AliDNS, DNSbyCommodo, zns-53 |

\* Indicates providers that potentially use Cloudflare backend service.

**Table 7: Name servers NXNAME deployment.**

standardized pseudo-type. Broader adoption of NXNAME could reduce ambiguity in *CDE* signaling and improve the interpretability of denial-of-existence responses.

**6.4.4 Adoption trend evolving since 2019.** We compare our results with prior work [15], which measured NSEC *CDE* and NSEC3 *White Lies* adoption in 2019. Notably, prior work did not examine NSEC *White Lies* or provide a unified view across NSEC/3 variations.

For .com, .net, and .org zones, we observe a substantial increase in the adoption of NSEC3 *White Lies*. For example, 18,879 apex domains utilized NSEC3 *White Lies* in 2019, whereas this number has increased to 468,991 in 2026, corresponding to 7% of apex domains that deploy NSEC3 in .com zone. Similarly, the adoption of NSEC *CDE* in the .com zone grew from 36,501 to 693,032 domains.

Our methodology enables a more comprehensive characterization of NSEC/3 variations, including previously unmeasured NSEC *White Lies* and their deployment patterns.

**6.4.5 Case study: Cloudflare.** We use Cloudflare as a representative case study to examine how real-world deployments align with documented implementations. Cloudflare is the dominant provider in NSEC deployments and exhibits highly consistent behavior across domains. Our observations closely match the behavior described in Cloudflare’s 2016 blog post [11], confirming the accuracy of our detection pipeline. In particular, Cloudflare consistently returns minimally covering NSEC responses with a single resource record and predominantly NODATA responses, which aligns with the characteristics of *CDE* identified in our analysis. More specifically, over 99.99% of domains hosted by Cloudflare return exactly one NSEC resource record, and over 99.97% return NODATA responses. In more than 99.99% of the collected responses, the owner name corresponds to the queried non-existent name, while the next owner name follows the pattern \000.{missingname}, consistent with the null-byte prefix strategy.

Beyond confirming prior behavior, we observe evidence of evolution in Cloudflare’s deployment. Specifically, over 90% of responses returned by Cloudflare include the NXNAME pseudo-type in the NSEC type bitmap, indicating alignment with the recent standardization in RFC 9824 [21].

**DNS Shotgun.** The remaining fraction of domains that do not include NXNAME likely correspond to wildcard synthesized responses. Among these domains, over 99.02% return 20 resource record types in the bitmap. This pattern aligns with the “DNS Shotgun” approach described by Cloudflare [11], where, for NODATA responses, the server includes a wide range of DNS record types except the queried type. This approach intends to reduce backend lookups and improve cost efficiency. However, as we discuss further in Section 8, this optimization introduces notable trade-offs, including response size

inflation and degraded semantic precision for Empty Non-Terminals (ENTs).

Overall, this case study demonstrates that our methodology accurately captures both the structural and semantic properties of real-world *CDE* deployments, while uncovering operational adoption of recent standards.

## 7 Wildcard in NSEC/3

While the previous section identifies NSEC/3 variants in the wild, wildcard synthesis represents another important yet under-measured dimension of DNSSEC denial-of-existence behavior.

Wildcards enable the dynamic synthesis of records for non-existent labels, and their interaction with DNSSEC denial-of-existence proofs introduces significant semantic complexity. In DNSSEC, negative responses are designed to provide cryptographic evidence that a name or a record type does not exist via NXDOMAIN (name error) and NODATA (type error), respectively. Wildcard synthesis changes this interpretation: a query for a non-existent name may trigger a synthesized match, effectively converting what would otherwise be an NXDOMAIN response into a positive response or a NODATA response.

Furthermore, wildcard applicability depends on the “closest encloser” rule; a wildcard is applied only when no closer existing name blocks the match, meaning that wildcard behavior can vary across different parts of the same zone. This interaction complicates the interpretation of negative responses, making their semantics dependent on context, zone structure, and server behavior.

In this section, we investigate wildcard synthesis in NSEC/3 deployments by analyzing response behaviors across different levels of the domain hierarchy. We first identify apex domains that potentially exhibit wildcard behavior, and then conduct a comparative analysis between these apex-level results and our WWW subdomain probing. By observing how response semantics transition from the apex to deeper labels, we characterize the interaction between wildcard expansion and denial-of-existence mechanism.

### 7.1 Wildcard Identification

We begin by identifying potential wildcard presence at the apex in NSEC/3 deployments, as illustrated in Figure 5.

We first classify responses based on their response codes into NXDOMAIN and NODATA. We exclude NXDOMAIN responses, as they indicate the non-existence of the queried name and thus the absence of wildcard synthesis at that label. For *CDE* deployments, the presence of the NXNAME pseudo-type similarly provides an explicit signal that the queried name does not exist, and therefore no wildcard synthesis is involved.

For the remaining NODATA responses, we determine whether wildcard synthesis is explicitly denied by examining the accompanying NSEC or NSEC3. For NSEC responses, we check whether the wildcard name (e.g., “\*.nsec-noext.com”) falls within any interval defined by the owner name and next owner name. If the wildcard is covered by such an interval, the response proves the non-existence of the wildcard. For NSEC3, we compute the hash of the wildcard domain using the parameters provided in the response and check whether it is covered by any returned NSEC3 interval. If so, the response similarly proves the non-existence of the wildcard.

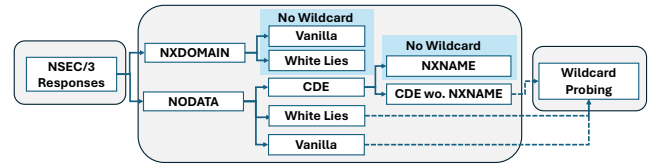


Figure 5: Identify wildcard synthesis apex. Dotted line suggests potential wildcard deployment.

After filtering out responses where wildcard non-existence is explicitly proven, the remaining cases correspond to zones where wildcard synthesis may be present.

**Prevalence of wildcard.** Table 8 summarizes the prevalence of potential wildcard behavior in our datasets. For both NSEC and NSEC3 data, we first evaluate whether records “explicitly acknowledge” a wildcard’s existence by identifying the wildcard label (e.g., “\*.nsec-noext.com”) or its corresponding hash in either the owner or next-owner name fields. In NSEC configurations, the appearance of a wildcard typically functions as part of a proof of the wildcard’s existence, establishing that no other records exist alphabetically between the predecessor and the wildcard. We also identify cases of “explicit denial,” where a negative response proves the non-existence of a wildcard by returning a record whose range covers the wildcard’s alphabetical or hashed position.

Our measurements reveal that about 11% to 14% of NSEC and around 7% of NSEC3 records acknowledge the existence of a wildcard in TLD zone datasets. The rate is lower for NSEC deployments in popularity-based datasets, likely due to the wild adoption of *CDE* (Table 6).

In contrast, explicit wildcard denial is rare across all datasets. This suggests that, for the remaining domains, authoritative name servers may rely on online-signing techniques which synthesize minimally covering responses that implicitly deny wildcard matches without requiring a dedicated wildcard-denial proof.

Overall, we find that at least 19% of NSEC-signed domains and 15% of NSEC3-signed domains in the TLD zone datasets potentially involve wildcard configurations. In the popularity-based datasets, these rates differ, with over 20% of NSEC domains and more than 10% of NSEC3 domains showing potential wildcard-related behavior. We use this subset for the subsequent analysis.

### 7.2 Uncovering Wildcard Zone Behavior

We combine the apex-level dataset with our WWW subdomain dataset to analyze wildcard behavior across different levels of the domain hierarchy. Table 11 in Appendix E summarizes the coverage of this comparison. We observe that about 15% of candidate domains do not return NSEC/3 responses at the subdomain level, suggesting that DNSSEC denial-of-existence behavior is not consistently applied across different levels of the hierarchy. For the remaining domains, we conduct a comparative analysis using domains that appear in both datasets.

**NSEC/3 variations.** Table 12 in Appendix E presents the proportion of *White Lies* and *CDE* for wildcard domains at the WWW subdomain level. For NSEC deployments, the proportions of *White Lies* and

|          | NSEC          |               |        |            |        | NSEC3         |               |        |            |        |
|----------|---------------|---------------|--------|------------|--------|---------------|---------------|--------|------------|--------|
|          | Deny wildcard | ACK. wildcard | %      | Candidates | %      | Deny wildcard | ACK. wildcard | %      | Candidates | %      |
| .com     | 1878          | 227716        | 14.44% | 325909     | 20.67% | 2698          | 453649        | 6.75%  | 729193     | 10.85% |
| .net     | 212           | 18642         | 11.61% | 33940      | 21.14% | 247           | 39005         | 7.18%  | 63230      | 11.64% |
| .org     | 202           | 17723         | 12.11% | 28985      | 19.81% | 169           | 39130         | 7.03%  | 58701      | 10.55% |
| Umbrella | 40            | 671           | 4.13%  | 4554       | 28.02% | 16            | 1157          | 12.15% | 1666       | 17.49% |
| Tranco   | 81            | 1883          | 3.84%  | 10232      | 20.85% | 16            | 5250          | 17.44% | 6866       | 22.80% |
| Radar    | 111           | 2791          | 4.29%  | 14686      | 22.58% | 15            | 7241          | 19.98% | 9598       | 26.48% |
| Majestic | 72            | 2256          | 5.50%  | 8722       | 21.28% | 4             | 5886          | 17.16% | 7835       | 22.85% |

**Table 8: Prevalence of wildcard in source domains. The percentages are computed against the total number of NSEC and NSEC3 apex, respectively; Candidates suggest the domains that potentially involve wildcard configurations.**

|          | NSEC                     |                        |               | NSEC3  |               |
|----------|--------------------------|------------------------|---------------|--------|---------------|
|          | NODATA<br>w.o.<br>NXNAME | NODATA<br>w.<br>NXNAME | NXDOMAIN      | NODATA | NXDOMAIN      |
| .com     | 49.34%                   | <b>13.41%</b>          | <b>37.24%</b> | 55.04% | <b>44.96%</b> |
| .net     | 55.63%                   | <b>16.87%</b>          | <b>27.51%</b> | 60.74% | <b>39.26%</b> |
| .org     | 52.33%                   | <b>15.98%</b>          | <b>31.68%</b> | 54.58% | <b>45.42%</b> |
| Umbrella | 63.57%                   | <b>28.67%</b>          | <b>7.76%</b>  | 63.63% | <b>36.37%</b> |
| Tranco   | 56.50%                   | <b>31.77%</b>          | <b>11.73%</b> | 67.38% | <b>32.62%</b> |
| Radar    | 57.32%                   | <b>31.61%</b>          | <b>11.07%</b> | 64.58% | <b>35.42%</b> |
| Majestic | 50.36%                   | <b>31.43%</b>          | <b>18.21%</b> | 69.25% | <b>30.75%</b> |

**Table 9: Distribution of NODATA and NXDOMAIN responses in the WWW dataset. For NSEC, NODATA responses are further classified based on the presence of NXNAME.**

*CDE* remain largely stable between apex and subdomain measurements, suggesting that these strategies are primarily determined by provider configuration (rather than by the queried hierarchy level). In contrast, NSEC3 deployments show an increase in *White Lies* at the WWW subdomain level across zone datasets. This indicates that queries beneath wildcard-enabled zones are more likely to trigger minimally covering denial responses, reflecting a shift toward privacy-preserving behavior.

**7.2.1 Shift from NODATA to NXDOMAIN.** Table 9 summarizes the distribution of NODATA and NXDOMAIN responses observed across NSEC/3 deployments in the WWW subdomain dataset. For NSEC, we further distinguish NODATA responses based on the presence of the NXNAME pseudo-type, which serves as an explicit NXDOMAIN signal in *CDE* deployments.

We observe that a substantial fraction of subdomain queries, 40%–50% in TLD zone datasets and 30%–40% in popularity-based datasets, result in either NXDOMAIN or NODATA responses with NXNAME. This is notable because all corresponding apex-level queries were previously classified as NODATA responses without NXNAME, indicating potential wildcard presence at the apex.

This shift can be explained by the *Closest Encloser* rule, as described in RFC 4592 [24]. A wildcard (e.g., “\*.nsec-noext.com”) is only applied when no closer matching label exists in the zone. When an explicit subdomain such as “sub.nsec-noext.com” is present, it becomes the closest encloser for queries beneath it (e.g., “non-ext.sub.nsec-noext.com”). As a result, the apex-level wildcard no longer applies, and the query may yield an NXDOMAIN response instead of triggering wildcard synthesis.

This observation highlights that the operational applicability of wildcard records is constrained by the internal structure of the zone.

In practice, domains often deploy explicit subdomains (e.g., *www*, *mail*) alongside wildcard records, and these nodes act as boundaries that suppress wildcard expansion. More broadly, this pattern reveals three signals: (1) the presence of wildcard behavior at the apex, (2) the existence of intermediate subdomains, and (3) the absence of effective wildcard expansion beneath those subdomains. Together, these signals allow us to distinguish between wildcard *presence* and its effective *coverage*, and to infer structural properties of the zone from negative response behavior.

**Security Considerations.** The observed transition from NODATA to NXDOMAIN should be interpreted primarily as a source of limited structural information rather than as a direct zone-enumeration capability. In particular, changes in wildcard synthesis behavior can indicate that an explicit intermediate name exists and acts as a closest encloser, thereby constraining the effective coverage of an apex-level wildcard. This allows an observer to infer certain properties of the zone structure from DNS response behavior. However, the technique does not directly enumerate zone contents and should not be considered a zone-walking attack. Moreover, some of the inferred names or structural information may already be observable through other public data sources, such as Certificate Transparency logs. Thus, the main contribution of this measurement is to characterize how wildcard behavior reveals boundaries in wildcard applicability and provides an additional signal about zone structure, rather than to demonstrate a new mechanism for comprehensive DNS name discovery.

**Takeaways.** We observe that 30% to 50% of domains shift from NODATA to NXDOMAIN for lower-depth queries, reveals limited structural information about the zone.

## 8 Discussion and Limitation

**DNS Shotgun.** We observe that Cloudflare adopts a *DNS Shotgun* strategy, in which the NSEC type bitmap includes a broad set of resource record types while excluding only the queried type. This behavior is consistent with prior descriptions of Cloudflare’s *CDE* implementation [11] and improves efficiency by avoiding additional backend look-ups for NODATA responses. However, this design introduces several operational and semantic trade-offs. First, the broad bitmap can obscure the distinction between a nonexistent name and an empty non-terminal, an ambiguity that NXNAME was standardized to address. Second, it inflates response size by including a larger NSEC Type Bitmaps field than necessary, partially conflicting with *CDE*’s goal of minimizing bytes on the wire. Finally, because the

bitmap may reflect provider-supported record types rather than the types actually present at the queried name, it reduces the semantic precision of authenticated denial and may mislead validators and type-inference tools. Some of these limitations are also discussed in Appendix A of RFC 9824 [21].

**NXNAME Adoption.** The NXNAME pseudo-type was introduced to explicitly signal compact denial of existence (CDE) [21], yet its adoption remains inconsistent. We find that several providers, including Route53, Azure, EuroDNS, and ALiDNS, implement CDE-like behavior without including NXNAME in their NSEC responses. The absence of NXNAME introduces ambiguity in interpreting denial semantics, complicating both client behavior and measurement inference. This suggests that the benefits of NXNAME are not fully realized in practice, and broader adoption would improve the clarity and consistency of DNSSEC signaling.

**Dataset and Measurement Artifacts.** Our analysis primarily focuses on the .com, .net, and .org TLDs, together with popularity-based domain datasets. This scope may introduce geographic and deployment biases, as we do not include country-code TLDs (ccTLDs), whose operational practices and DNS configurations may differ across regions. We use OpenINTEL as our primary data sources; while it has been widely used in many prior measurement studies [1, 4, 23, 38], its coverage and vantage points may still influence the representativeness of our results.

Our measurements rely on AFSDB queries. Because AFSDB is deprecated and rarely used, some resolvers or middle-boxes could filter or handle such queries differently, making it a potential source of measurement artifacts. At the same time, its rarity is precisely what makes it useful for eliciting NODATA responses: for a standards-compliant DNSSEC deployment, querying an existing but typically absent record type should produce a well-defined negative response rather than application-specific behavior. In contrast, commonly used types such as A may trigger resolver- or provider-specific synthesis, which can complicate interpretation of denial-of-existence behavior. We do not observe evidence of widespread AFSDB filtering in our measurements, but we treat its use as a limitation when interpreting the results.

**Other Limitations.** Our measurements rely on a single recursive DNS resolver, which may introduce resolver-specific artifacts, including effects from caching behavior and differences in how DNS responses are processed or returned. Our wildcard analysis focuses specifically on WWW subdomains; because other subdomains may be configured, operated, or managed differently, the results may not generalize to wildcard behavior across an entire domain. Finally, our analysis is based on snapshot measurements and therefore captures configurations only at particular points in time. DNS deployments may evolve due to operational changes, migrations, or policy updates, so the observed configurations may not reflect behavior over longer periods. A longitudinal analysis across multiple measurement periods would help quantify such temporal variation and is an important direction for future work.

## 9 Conclusion

We present a large-scale measurement study of authenticated denial-of-existence in DNSSEC deployments, covering both NSEC and NSEC3, and their operational variants. We develop a methodology

to distinguish denial mechanisms in the wild and characterize how providers implement minimally covering strategies in practice. Our analysis highlights the diversity in real-world NSEC/3 deployment, including the prevalence of *minimally covering NSEC*, *White Lies*, and *CDE* variants. We further show that wildcard synthesis is non-negligible and that its interaction with denial-of-existence proofs lead to non-uniform wildcard coverage across the domain hierarchy. These behaviors can reveal limited structural information about DNS zones, such as boundaries in wildcard applicability, without directly enabling zone enumeration.

## Acknowledgments

We thank anonymous reviewers for their insightful and constructive suggestions and feedback. This work is supported by National Science Foundation CNS-2319421 and the Commonwealth Cyber Initiative. Additionally, this work used AWS via Cloudbank [28] through allocation CIS260181 from the Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support (ACCESS) program [8], which is supported by U.S. National Science Foundation grants #2138259, #2138286, #2138307, #2137603, and #2138296. Finally, the research leading to these results was made possible by OpenINTEL (<https://www.openintel.nl/>), a joint project of the University of Twente, SIDN, NLnet Labs, and SURF.

## References

- [1] Abhishta Abhishta, Roland van Rijswijk-Deij, and Lambert JM Nieuwenhuis. 2019. Measuring the impact of a successful DDoS attack on the customer behaviour of managed DNS service providers. *ACM SIGCOMM Computer Communication Review* 48, 5 (2019), 70–76.
- [2] Russ Allbery. 2010. DNS SRV Resource Records for AFS. RFC 5864. doi:10.17487/RFC5864
- [3] Roy Arends, Geoffrey Sisson, David Blacka, and Ben Laurie. 2008. DNS Security (DNSSEC) Hashed Authenticated Denial of Existence. RFC 5155. doi:10.17487/RFC5155
- [4] Alfred Arouna, Rick Fontein, Bernd Meijerink, Ioana Livadariu, and Mattijs Jonker. 2025. On the Role of Forward-Confirmed reverse DNS in E-mail Authentication. In *2025 9th Network Traffic Measurement and Analysis Conference (TMA)*. IEEE, 1–11.
- [5] Md Ishtiaq Ashiq, Olivier Hureau, Casey Deccio, and Tijay Chung. 2025. Decoding DNSSEC Errors at Scale: An Automated DNSSEC Error Resolution Framework using Insights from DNSViz Logs. In *Proceedings of the 2025 ACM Internet Measurement Conference*. 242–257.
- [6] AWS. 2026. DNSSEC proofs of nonexistence in Route 53. <https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/dns-configuring-dnssec-proof-of-nonexistence.html> (Accessed: Apr. 24, 2026).
- [7] Steven M Bellovin. 1995. Using the Domain Name System for System Break-ins.. In *USENIX Security Symposium*.
- [8] Timothy J Boerner, Stephen Deems, Thomas R Furlani, Shelley L Knuth, and John Towns. 2023. Access: Advancing innovation: NSF's advanced cyberinfrastructure coordination ecosystem: Services & support. In *Practice and experience in advanced research computing 2023: Computing for the common good*. 173–176.
- [9] Taejoong Chung, Roland van Rijswijk-Deij, Balakrishnan Chandrasekaran, David Choffnes, Dave Levin, Bruce M. Maggs, Alan Mislove, and Christo Wilson. 2017. A Longitudinal, End-to-End View of the DNSSEC Ecosystem. In *26th USENIX Security Symposium (USENIX Security 17)*. USENIX Association, Vancouver, BC, 1307–1322. <https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/chung>
- [10] Cisco. 2025. Umbrella Popularity List. <https://umbrella-static.s3-us-west-1.amazonaws.com/index.html> (Accessed: Apr. 19, 2026).
- [11] Cloudflare. 2016. Economical With The Truth: Making DNSSEC Answers Cheap. <https://blog.cloudflare.com/black-lies/> (Accessed: Apr. 24, 2026).
- [12] Cloudflare Radar. 2025. Domain Rankings. <https://radar.cloudflare.com/domains> (Accessed: Apr. 19, 2026).
- [13] Cordian Alexander Daniluk, Yevheniya Nosyk, Andrzej Duda, and Maciej Korczyński. 2024. Zeros Are Heroes: NSEC3 Parameter Settings in the Wild. In *Proceedings of the 2024 ACM on Internet Measurement Conference*. 415–422.
- [14] Casey Deccio and Benjamin Tessem. 2025. On Aggressive Negative Caching in DNS Resolvers. In *2025 9th Network Traffic Measurement and Analysis Conference*

- (TMA). IEEE, 1–9.
- [15] Jonathan Demke and Casey Deccio. 2019. On DNSSEC Negative Responses, Lies, and Zone Size Detection. In *International Conference on Passive and Active Network Measurement*. Springer, 231–243.
  - [16] David Dittrich, Erin Kenneally, et al. 2012. *The Menlo Report: Ethical principles guiding information and communication technology research*. Technical Report. US Department of Homeland Security.
  - [17] R. (Miek) Gieben and Matthijs Mekking. 2014. Authenticated Denial of Existence in the DNS. RFC 7129. doi:10.17487/RFC7129
  - [18] Olivia Gruza, Elias Heftrig, Oliver Jacobsen, Haya Schulmann, Niklas Vogel, and Michael Waidner. 2024. Attacking with Something That Does Not Exist: 'Proof of {Non-Existence}' Can Exhaust {DNS} Resolver {CPU}. In *18th USENIX WOOT Conference on Offensive Technologies (WOOT 2024)*. 45–57.
  - [19] Wes Hardaker and Viktor Dukhovni. 2022. Guidance for NSEC3 Parameter Settings. RFC 9276. doi:10.17487/RFC9276
  - [20] Shumon Huque. 2025. A Survey of Authenticated Denial of Existence in DNSSEC. DNS-OARC 44. <https://indico.dns-oarc.net/event/52/contributions/1147/attachments/1103/2287/AuthenticatedDenial.pdf> (Accessed: Apr. 13, 2026).
  - [21] S. Huque, C. Elmerot, and O. Gudmundsson. 2025. Compact Denial of Existence in DNSSEC. RFC 9824. doi:10.17487/RFC9824
  - [22] Dan Kaminsky. 2008. It's The End Of The Cache As We Know It. Black Hat 2008. <https://db.ci/wp-content/uploads/2021/10/dmkbo2k8-101127182039-phpapp02.pdf> (Accessed: Apr. 13, 2026).
  - [23] Calvin Kranig, Eric Pauley, Wei-Shiang Wung, Paul Barford, Mark Crovella, and Joel Sommers. 2025. Toward a Representative DNS Data Corpus: A Longitudinal Comparison of Collection Methods. In *2025 9th Network Traffic Measurement and Analysis Conference (TMA)*. 1–11. doi:10.23919/TMA66427.2025.11096967
  - [24] Edward P. Lewis. 2006. The Role of Wildcards in the Domain Name System. RFC 4592. doi:10.17487/RFC4592
  - [25] Majestic. 2025. Majestic Million. <https://majestic.com/reports/majestic-million> (Accessed: Apr. 19, 2026).
  - [26] Microsoft Ignite. 2026. DNSSEC overview. <https://learn.microsoft.com/en-us/azure/dns/dnssec> (Accessed: Aug. 24, 2026).
  - [27] Frederic P Miller, Agnes F Vandome, and John McBrewhster. 2009. Levenshtein distance: Information theory, computer science, string (computer science), string metric, damerau? levenshtein distance, spell checker, hamming distance.
  - [28] Michael Norman, Vince Kellen, Shava Smallen, Brian DeMeulle, Shawn Strande, Ed Lazowska, Naomi Alterman, Rob Fatland, Sarah Stone, Amanda Tan, et al. 2021. Cloudbank: Managed services to simplify cloud access for computer science research and education. In *Practice and Experience in Advanced Research Computing 2021: Evolution Across All Dimensions*. 1–4.
  - [29] OpenINTEL. 2026. Forward DNS Measurements: Top Lists. <https://openintel.nl/data/forward-dns/top-lists/> (Accessed: Apr. 24, 2026).
  - [30] OpenINTEL. 2026. OpenINTEL: Active DNS Measurement Project. <https://openintel.nl/> (Accessed: Apr. 15, 2026).
  - [31] Paul Pearce, Ben Jones, Frank Li, Roya Ensafi, Nick Feamster, Nick Weaver, and Vern Paxson. 2017. Global Measurement of DNS Manipulation. In *26th USENIX Security Symposium (USENIX Security 17)*. 307–323.
  - [32] Victor Le Pochat, Tom Van Goethem, Samaneh Tajalizadehkhoob, Maciej Koczynski, and Wouter Joosen. 2025. Tranco: A Research-Oriented Top Sites Ranking Hardened Against Manipulation. <https://tranco-list.eu/> (Accessed: Apr. 19, 2026).
  - [33] Scott Rose, Matt Larson, Dan Massey, Rob Austein, and Roy Arends. 2005. DNS Security Introduction and Requirements. RFC 4033. doi:10.17487/RFC4033
  - [34] Scott Rose, Matt Larson, Dan Massey, Rob Austein, and Roy Arends. 2005. Protocol Modifications for the DNS Security Extensions. RFC 4035. doi:10.17487/RFC4035
  - [35] Scott Rose, Matt Larson, Dan Massey, Rob Austein, and Roy Arends. 2005. Resource Records for the DNS Security Extensions. RFC 4034. doi:10.17487/RFC4034
  - [36] Joe St Sauver. 2017. Zone Walking (Zone Enumeration via DNSSEC NSEC Records). <https://www.domaintools.com/blog/zone-walking-zone-enumeration-via-dnssec-nsec-records> (Accessed: Apr. 18, 2026).
  - [37] Shumon Huque. 2026. Analysis of UltraDNS Minimally Covering NSEC Algorithm. [https://github.com/shuque/minimal\\_nsec/blob/main/UltraDNS.md](https://github.com/shuque/minimal_nsec/blob/main/UltraDNS.md) (Accessed: Apr. 24, 2026).
  - [38] Raffaele Sommes, Gautam Akiwate, Mattijs Jonker, Giovane C Moura, Marco Davids, Roland van Rijswijk-Deij, Geoffrey M Voelker, Stefan Savage, Anna Sperotto, et al. 2021. Characterization of anycast adoption in the DNS authoritative infrastructure. In *Network Traffic Measurement and Analysis Conference (TMA'21)*.
  - [39] Roland van Rijswijk-Deij, Mattijs Jonker, Anna Sperotto, and Aiko Pras. 2016. A High-Performance, Scalable Infrastructure for Large-Scale Active DNS Measurements. *IEEE Journal on Selected Areas in Communications* 34, 6 (2016), 1877–1888. doi:10.1109/JSAC.2016.2558918
  - [40] Sam Weiler and Johan Stenstam. 2006. Minimally Covering NSEC Records and DNSSEC On-line Signing. RFC 4470. doi:10.17487/RFC4470
  - [41] Yizhe Zhang and Hyeonmin Lee and Shumon Huque and Yixin Sun. 2026. Open-source repository accompanying this paper. <https://github.com/UVA-NetSec/dnsssec26> (Accessed: Aug. 24, 2026).

|          | Timeout | CNAME  | Only SOA |
|----------|---------|--------|----------|
| .com     | 8.87%   | 22.93% | 88.66%   |
| .net     | 9.92%   | 39.23% | 75.19%   |
| .org     | 15.88%  | 31.39% | 73.91%   |
| Umbrella | 58.35%  | 20.97% | 41.65%   |
| Tranco   | 28.38%  | 35.81% | 71.40%   |
| Radar    | 21.49%  | 47.76% | 78.21%   |
| Majestic | 41.49%  | 30.03% | 57.99%   |

**Table 10: Categorization of domains with DNSKEY but without NSEC3.**

## A Ethics

Our study was conducted in strict accordance with established research ethics and the Menlo Report principles [16].

**Stakeholders and Ethical Impact.** Our measurement involves authoritative nameserver operators, public DNS service providers, and domain owners whose infrastructure may respond to our probes. Because our study focuses on DNSSEC denial-of-existence behavior, our measurements do not target end users or collect any personally identifiable information. We only issue standard DNS queries that conform to normal protocol behavior and resemble routine negative lookups commonly observed in operational environments. The measurements are designed solely to observe protocol behavior and deployment characteristics.

**Data Collection.** Our analysis primarily relies on OpenINTEL [30, 39], an established DNS measurement platform that follows responsible measurement practices. For our active probing, we perform additional AFSDb queries for non-existent labels to elicit denial-of-existence responses. To reduce operational impact, we only probe DNSSEC-enabled apex domains identified by our data loader module and limit measurements to a small number of queries per domain. Queries are performed at a controlled rate and are not repeated aggressively. Finally, we store only protocol-level DNS response data necessary for analysis, including DNS resource records and response codes. When presenting examples in the paper, we anonymize domain names (i.e., “nsec-noext.com”) where appropriate to avoid unnecessary exposure of operational details.

## B Domains with DNSKEY but without NSEC/3

We identify several underlying reasons behind Domains with DNSKEY but without NSEC/3, as summarized in Table 10: (i) a set of domains is attributable to name server errors or timeouts, (ii) another portion involves delegations via CNAME records where the target zones are not DNSSEC-signed, and (iii) additionally, we observe domains return only SOA records and domains exhibit completely empty authority sections. These results suggest that the absence of NSEC/3 responses in otherwise signed domains is often due to operational delegation or incomplete DNSSEC deployments.

## C Name servers

Table 13 listed the top 5 dominant name servers across zone and popularity based datasets in both OpenINTEL and our WWW probing. Cloudflare strongly dominates NSEC deployments, while NSEC3 is

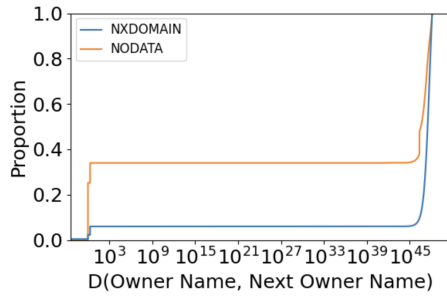


Figure 6: Distance between owner name and next owner name for NSEC3 responses (log-scale).

|          | Wildcard Candidate | Observed in WWW | %      |
|----------|--------------------|-----------------|--------|
| .com     | 1,054,354          | 886,770         | 84.11% |
| .net     | 97,074             | 84,790          | 87.35% |
| .org     | 87,634             | 75,844          | 86.55% |
| Umbrella | 6,196              | 4,953           | 79.94% |
| Tranco   | 17,054             | 15,909          | 93.29% |
| Radar    | 24,223             | 22,926          | 94.65% |
| Majestic | 16,518             | 15,601          | 94.45% |

Table 11: WWW coverage of wildcard candidates.

more distributed across providers such as Google Domains, Wix, and OVHCloud. The provider distributions for source and WWW domains are generally similar, suggesting that the dominant deployment patterns persist across these two levels of the domain hierarchy.

## D DNS Canonical Distance

Figure 6 illustrates the distribution of NSEC3 interval distances in hash space on a logarithmic scale. We observe a clear disparity between domains that employ minimally covering strategies and those that do not. In the .com zone, approximately 33.97% of NODATA records and 5.96% of NXDOMAIN records exhibit very small distances ( $D \leq 2$ ), indicating the use of minimally covering intervals. In contrast, the remaining records show substantially larger distances, often spanning several orders of magnitude.

## E Wildcard Characterization

**Wildcard coverage.** Table 11 summarizes the number of domains identified as wildcard candidates at the apex and the subset that also exhibit denial-of-existence behavior when querying the none xistent.www subdomain. We observe that approximately 15% of domains do not return NSEC/3 responses at the subdomain level, suggesting that DNSSEC deployment is not consistently applied across different levels of the zone.

**NSEC/3 variations in WWW and source datasets.** Table 12 summarizes the NSEC/3 variations across the WWW and source datasets. We do not observe substantial changes in the distribution of NSEC deployments. For NSEC3, the proportion of *White Lies* is slightly higher at the WWW subdomain level compared to the overall proportion at the apex level.

|        |          | NSEC              |            | NSEC3             |            |
|--------|----------|-------------------|------------|-------------------|------------|
|        |          | <i>White Lies</i> | <i>CDE</i> | <i>White Lies</i> | <i>CDE</i> |
| WWW    | .com     | 0.85%             | 32.20%     | 43.23%            | 0.00%      |
|        | .net     | 1.00%             | 47.30%     | 38.50%            | 0.00%      |
|        | .org     | 0.96%             | 41.05%     | 46.03%            | 0.00%      |
|        | Umbrella | 1.08%             | 85.19%     | 19.85%            | 0.00%      |
|        | Tranco   | 0.81%             | 81.16%     | 22.86%            | 0.00%      |
|        | Radar    | 0.58%             | 80.73%     | 22.86%            | 0.00%      |
|        | Majestic | 0.86%             | 73.26%     | 21.69%            | 0.00%      |
| Source | .com     | 0.85%             | 32.27%     | 38.43%            | 0.00%      |
|        | .net     | 1.00%             | 47.39%     | 34.69%            | 0.00%      |
|        | .org     | 0.95%             | 41.17%     | 37.29%            | 0.00%      |
|        | Umbrella | 1.05%             | 85.27%     | 18.86%            | 0.00%      |
|        | Tranco   | 0.81%             | 81.09%     | 15.41%            | 0.00%      |
|        | Radar    | 0.59%             | 80.61%     | 16.59%            | 0.00%      |
|        | Majestic | 0.86%             | 73.31%     | 20.81%            | 0.00%      |

Table 12: NSEC/3 variations for wildcard domains at WWW datasets.

| OpenINTEL         |        |               |        | www               |        |               |        |
|-------------------|--------|---------------|--------|-------------------|--------|---------------|--------|
| NSEC              |        | NSEC3         |        | NSEC              |        | NSEC3         |        |
| NS                | %      | NS            | %      | NS                | %      | NS            | %      |
| <b>.com</b>       |        |               |        |                   |        |               |        |
| cloudflare        | 38.03% | googledomains | 39.98% | cloudflare        | 36.13% | googledomains | 41.65% |
| infomaniak        | 7.44%  | wixdns        | 19.40% | infomaniak        | 7.94%  | wixdns        | 19.61% |
| registrar-servers | 5.82%  | ovh           | 9.60%  | registrar-servers | 6.17%  | spaceship     | 10.12% |
| domaincontrol     | 4.90%  | one           | 3.97%  | domaincontrol     | 5.17%  | ovh           | 9.74%  |
| register          | 4.73%  | openprovider  | 2.88%  | register          | 4.72%  | one           | 3.97%  |
| <b>.net</b>       |        |               |        |                   |        |               |        |
| cloudflare        | 50.01% | googledomains | 45.05% | cloudflare        | 50.98% | googledomains | 46.64% |
| infomaniak        | 5.48%  | wixdns        | 14.80% | infomaniak        | 5.61%  | wixdns        | 14.84% |
| registrar-servers | 4.81%  | ovh           | 9.89%  | registrar-servers | 4.89%  | ovh           | 9.96%  |
| register          | 3.88%  | one           | 3.83%  | register          | 3.69%  | spaceship     | 6.40%  |
| domaincontrol     | 2.86%  | openprovider  | 2.76%  | domaincontrol     | 2.88%  | one           | 3.82%  |
| <b>.org</b>       |        |               |        |                   |        |               |        |
| cloudflare        | 47.07% | googledomains | 46.74% | cloudflare        | 48.10% | googledomains | 48.16% |
| infomaniak        | 7.48%  | wixdns        | 15.48% | infomaniak        | 7.71%  | wixdns        | 15.51% |
| registrar-servers | 5.24%  | ovh           | 7.73%  | registrar-servers | 5.32%  | spaceship     | 10.21% |
| domaincontrol     | 4.24%  | one           | 3.08%  | domaincontrol     | 4.32%  | ovh           | 7.77%  |
| register          | 2.78%  | openprovider  | 2.24%  | register          | 2.57%  | one           | 3.07%  |
| <b>Umbrella</b>   |        |               |        |                   |        |               |        |
| cloudflare        | 56.59% | googledomains | 15.85% | cloudflare        | 62.83% | googledomains | 16.54% |
| awsdns            | 12.25% | d-53          | 11.93% | awsdns            | 10.26% | ovh           | 6.66%  |
| ripe              | 1.40%  | ovh           | 7.72%  | domaincontrol     | 1.43%  | akam          | 3.82%  |
| arin              | 1.40%  | akam          | 6.47%  | nsone             | 1.34%  | d-53          | 3.22%  |
| nsone             | 1.35%  | transip       | 3.18%  | azure-dns         | 1.13%  | timeweb       | 2.92%  |
| <b>Tranco</b>     |        |               |        |                   |        |               |        |
| cloudflare        | 62.69% | googledomains | 11.30% | cloudflare        | 64.75% | googledomains | 11.71% |
| awsdns            | 6.31%  | ovh           | 8.98%  | awsdns            | 6.47%  | ovh           | 8.49%  |
| dns               | 2.42%  | timeweb       | 5.30%  | domaincontrol     | 1.98%  | timeweb       | 5.11%  |
| domaincontrol     | 1.85%  | transip       | 5.19%  | timeweb           | 1.66%  | transip       | 4.88%  |
| timeweb           | 1.49%  | akam          | 4.43%  | azure-dns         | 1.06%  | spaceship     | 4.62%  |
| <b>Radar</b>      |        |               |        |                   |        |               |        |
| cloudflare        | 68.40% | googledomains | 12.68% | cloudflare        | 62.77% | googledomains | 12.82% |
| awsdns            | 5.52%  | ovh           | 10.11% | awsdns            | 5.02%  | ovh           | 10.74% |
| domaincontrol     | 1.77%  | transip       | 5.45%  | domaincontrol     | 1.62%  | transip       | 5.68%  |
| dns               | 1.54%  | timeweb       | 5.10%  | timeweb           | 1.14%  | timeweb       | 5.51%  |
| timeweb           | 1.16%  | one           | 3.34%  | registrar-servers | 1.05%  | spaceship     | 3.87%  |
| <b>Majestic</b>   |        |               |        |                   |        |               |        |
| cloudflare        | 61.43% | ovh           | 8.72%  | cloudflare        | 61.67% | ovh           | 9.40%  |
| awsdns            | 5.46%  | googledomains | 8.44%  | awsdns            | 5.38%  | googledomains | 8.75%  |
| domaincontrol     | 2.36%  | transip       | 7.14%  | domaincontrol     | 2.44%  | transip       | 7.71%  |
| timeweb           | 1.85%  | timeweb       | 6.76%  | timeweb           | 1.99%  | timeweb       | 7.36%  |
| infomaniak        | 1.42%  | one           | 3.46%  | infomaniak        | 1.54%  | spaceship     | 4.25%  |

Table 13: Top 5 dominant name servers in NSEC/3.